

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	58394	52.134707
1	[Git, Change, Create]	17242	15.393818
2	[Git, Commit]	11260	10.053033
3	[Git, Change, Delete]	8758	7.819224
4	[File, Git]	5891	5.259540
5	[Git, Change, Rename]	3354	2.994482
6	[Git, Tag]	1732	1.546346
7	[Author, Git, Person]	1265	1.129404
8	[Json, Key]	668	0.596397
9	[Json, Value, Scalar]	603	0.538364
10	[Committer, Git, Person]	370	0.330339
11	[NPM, Dependency]	338	0.301770
12	[Type, TS, Primitive]	293	0.261593
13	[Type, TS, Declared]	277	0.247308
14	[TS, ExternalDeclaration]	215	0.191954
15	[Git, Change, Copy]	138	0.123208
16	[Type, TS, Literal]	136	0.121422
17	[Json, Value, Object]	133	0.118744
18	[Type, TS, Union]	120	0.107137
19	[Type, TS, ObjectMember]	103	0.091959
20	[NPM, Script]	91	0.081246
21	[TS, Property]	65	0.058033
22	[TS, Function]	47	0.041962
23	[Git, Branch]	44	0.039284
24	[Type, TS, FunctionParameter]	40	0.035712
25	[Type, Object, TS]	39	0.034820
26	[File, Directory]	34	0.030356
27	[Type, TS, Function]	34	0.030356
28	[TS, Parameter]	33	0.029463
29	[Package, File, Json, NPM]	29	0.025891

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

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Nodes per label combination (more than 0.5% overall)

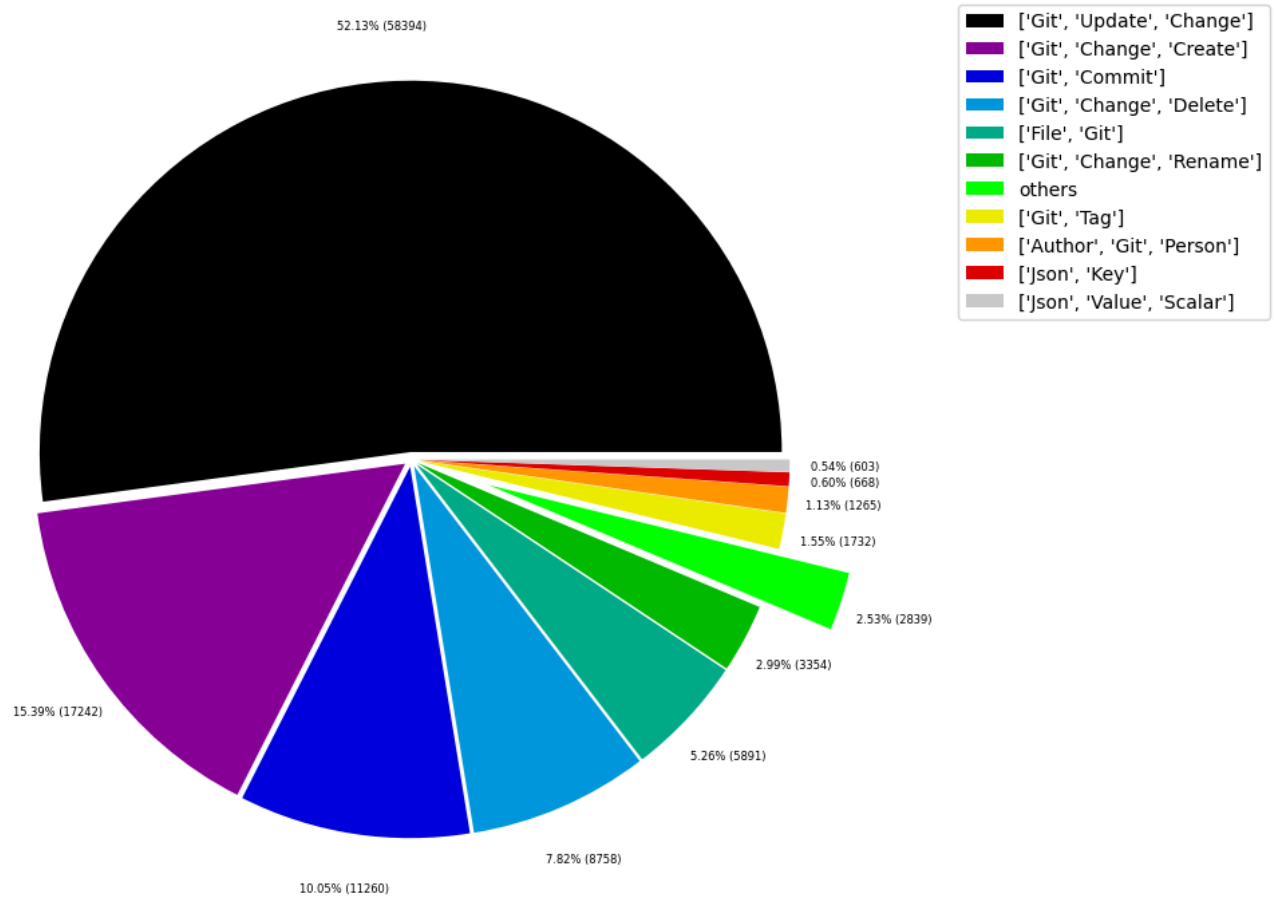


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000893
1	[NPM, PublishConfig]	1	0.000893
2	[File, TS, Scan]	1	0.000893
3	[TS, Method]	1	0.000893
4	[Repository, File, Git]	1	0.000893
5	[TS, Constructor]	1	0.000893
6	[Value, TS, ObjectMember]	1	0.000893
7	[TS, Class]	1	0.000893
8	[TS, Enum]	2	0.001786
9	[Value, Object, TS]	3	0.002678
10	[Type, TS, Tuple]	3	0.002678
11	[Value, TS, Function]	4	0.003571
12	[TS, TypeParameter]	4	0.003571
13	[Value, TS, Complex]	5	0.004464
14	[Repository, NPM]	5	0.004464
15	[NPM, Engine]	6	0.005357
16	[Project, TS]	6	0.005357
17	[File, Local]	6	0.005357
18	[Type, TS, TypeParameterReference]	6	0.005357
19	[Value, TS, Member]	6	0.005357
20	[File, TS, Local, Module]	6	0.005357
21	[Value, TS, Call]	6	0.005357
22	[TS, EnumMember]	8	0.007142
23	[Type, TS, NotIdentified]	11	0.009821
24	[TS, ExternalModule]	11	0.009821
25	[Json, Value, Array]	12	0.010714
26	[Value, TS, Declared]	13	0.011607
27	[TS, TypeAlias]	16	0.014285
28	[File, Directory, Local]	16	0.014285
29	[TS, Interface]	17	0.015178

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

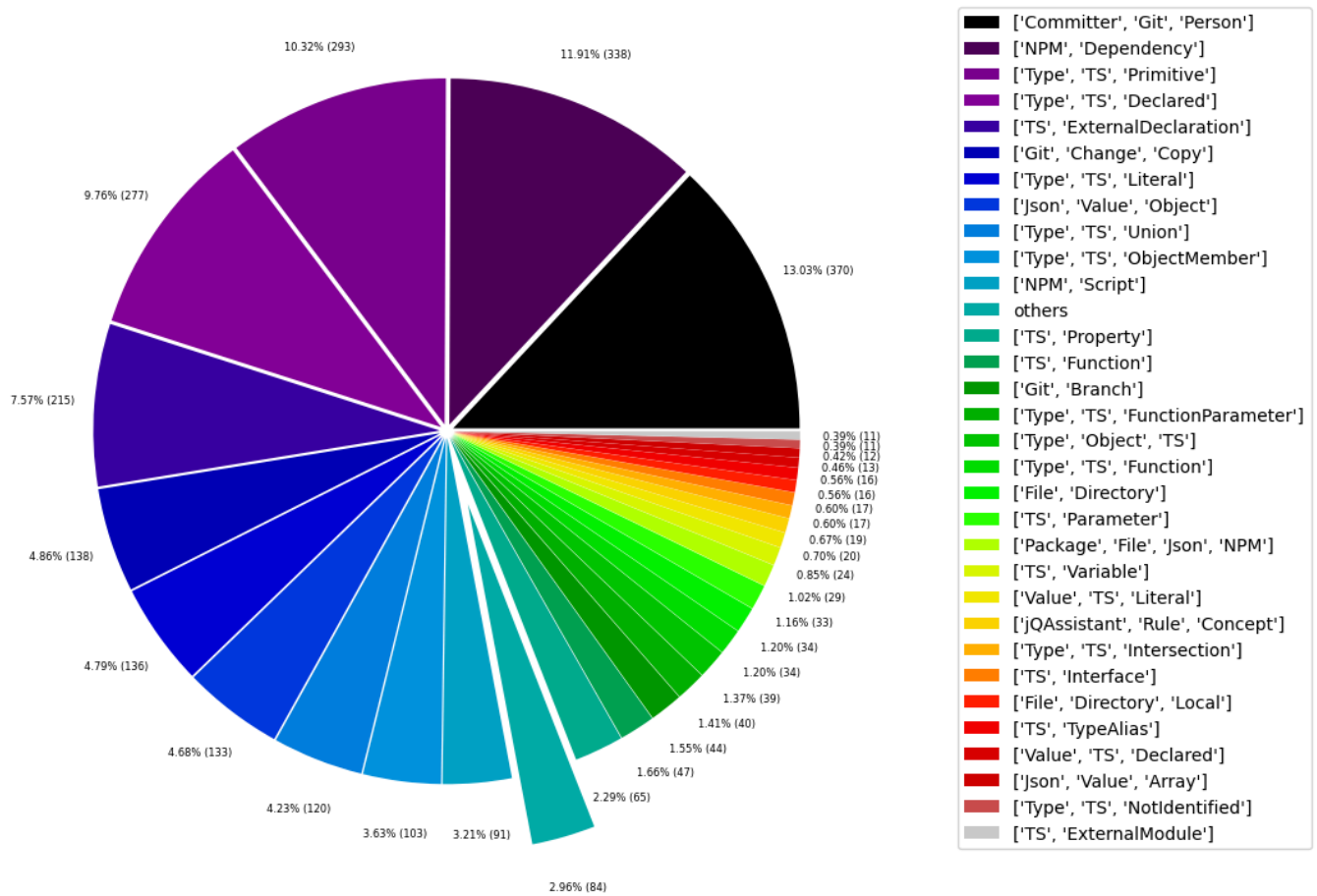


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

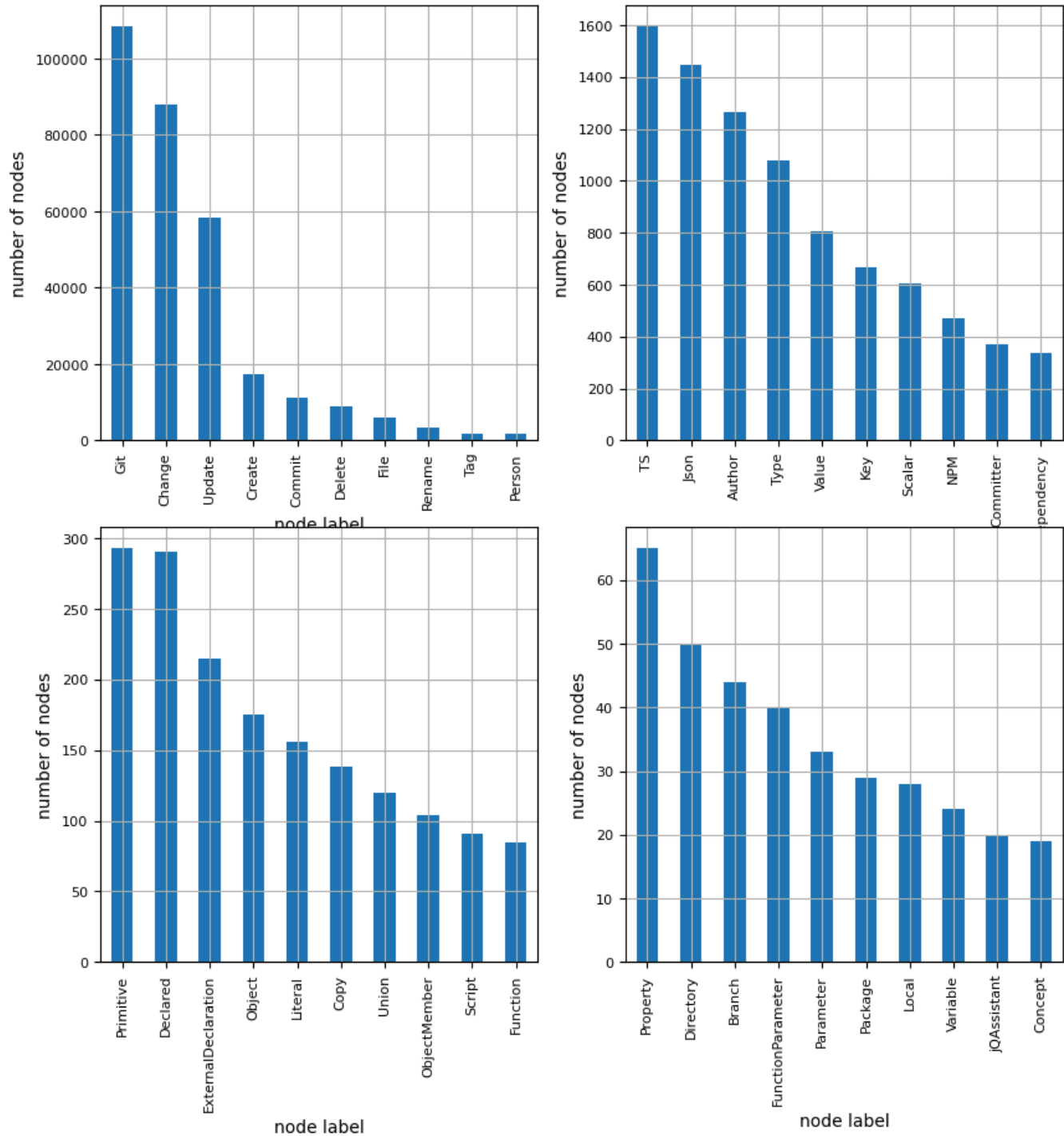
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	108449	96.824277
1	Change	87886	78.465439
2	Update	58394	52.134707
3	Create	17242	15.393818
4	Commit	11260	10.053033
5	Delete	8758	7.819224
6	File	5984	5.342571
7	Rename	3354	2.994482
8	Tag	1732	1.546346
9	Person	1635	1.459743
10	TS	1595	1.424031
11	Json	1445	1.290109
12	Author	1265	1.129404
13	Type	1079	0.963341
14	Value	806	0.719604
15	Key	668	0.596397
16	Scalar	603	0.538364
17	NPM	470	0.419620
18	Committer	370	0.330339
19	Dependency	338	0.301770
20	Primitive	293	0.261593
21	Declared	290	0.258915
22	ExternalDeclaration	215	0.191954
23	Object	175	0.156242
24	Literal	156	0.139278
25	Copy	138	0.123208
26	Union	120	0.107137
27	ObjectMember	104	0.092852
28	Script	91	0.081246
29	Function	85	0.075889
30	Property	65	0.058033
31	Directory	50	0.044640
32	Branch	44	0.039284
33	FunctionParameter	40	0.035712
34	Parameter	33	0.029463
35	Package	29	0.025891
36	Local	28	0.024999
37	Variable	24	0.021427
38	jqAssistant	20	0.017856
39	Concept	19	0.016963

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 334464

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	87886	26.276670
1	MODIFIES	87886	26.276670
2	UPDATES	58394	17.458979
3	COMMITTED	22520	6.733161
4	CREATES	20734	6.199172
5	HAS_PARENT	12340	3.689485
6	DELETES	12112	3.621316
7	HAS_COMMIT	11260	3.366581
8	HAS_FILE	5891	1.761326
9	RENAMES	3354	1.002799
10	HAS_NEW_NAME	1770	0.529205
11	HAS_TAG	1732	0.517843
12	ON_COMMIT	1732	0.517843
13	HAS_AUTHOR	1265	0.378217
14	DEPENDS_ON	887	0.265200
15	HAS_KEY	668	0.199723
16	HAS_VALUE	668	0.199723
17	CONTAINS	596	0.178196
18	HAS_COMMITTER	370	0.110625
19	OF_TYPE	339	0.101356
20	EXPORTS	309	0.092387
21	REFERENCES	197	0.058900
22	DECLARES	186	0.055611
23	DECLARES_DEV_DEPENDENCY	169	0.050529
24	DECLARES_DEPENDENCY	161	0.048137
25	COPIES	138	0.041260
26	HAS_MEMBER	104	0.031095
27	HAS_TYPE_ARGUMENT	94	0.028105
28	DECLARES_SCRIPT	91	0.027208
29	RETURNS	82	0.024517

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

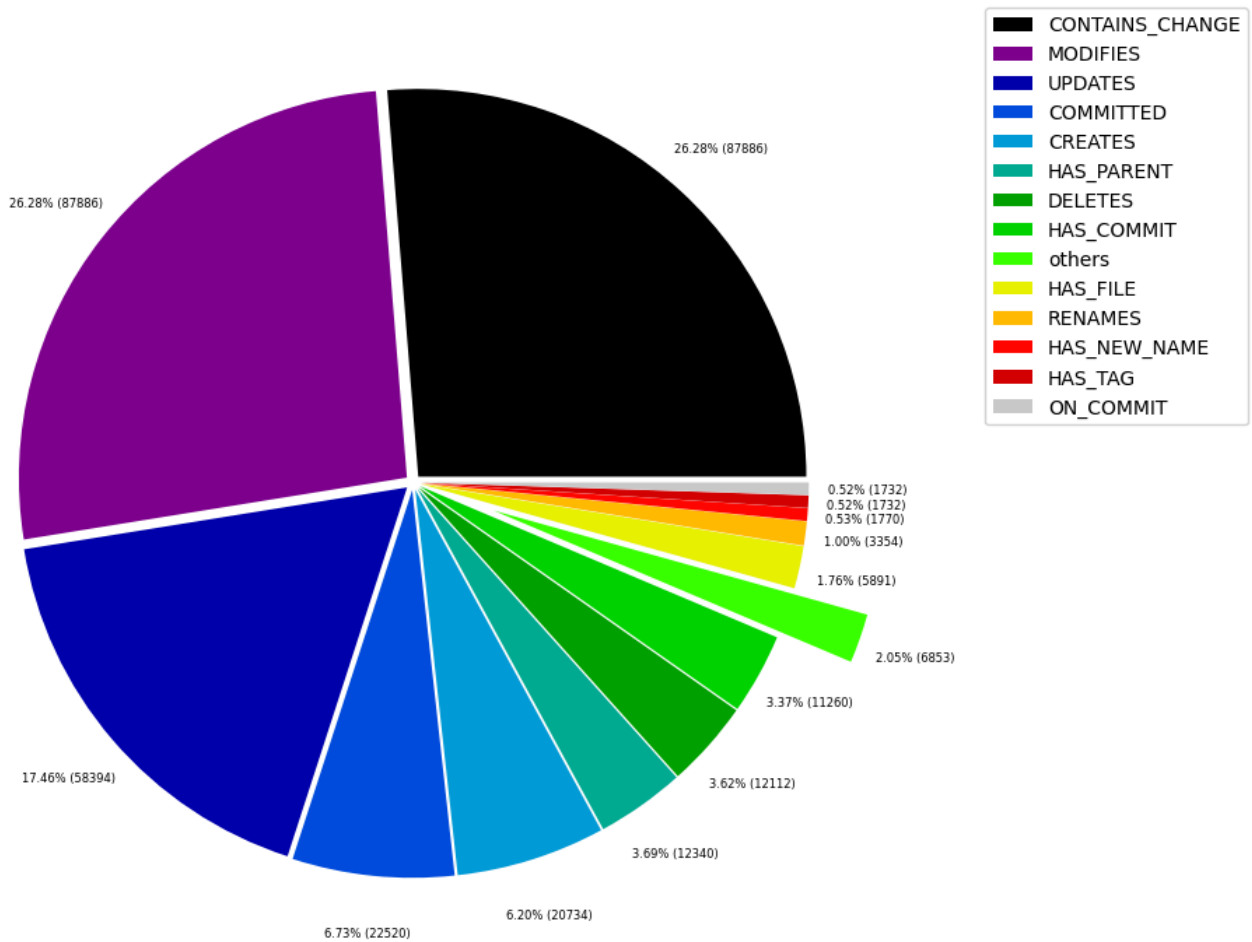


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

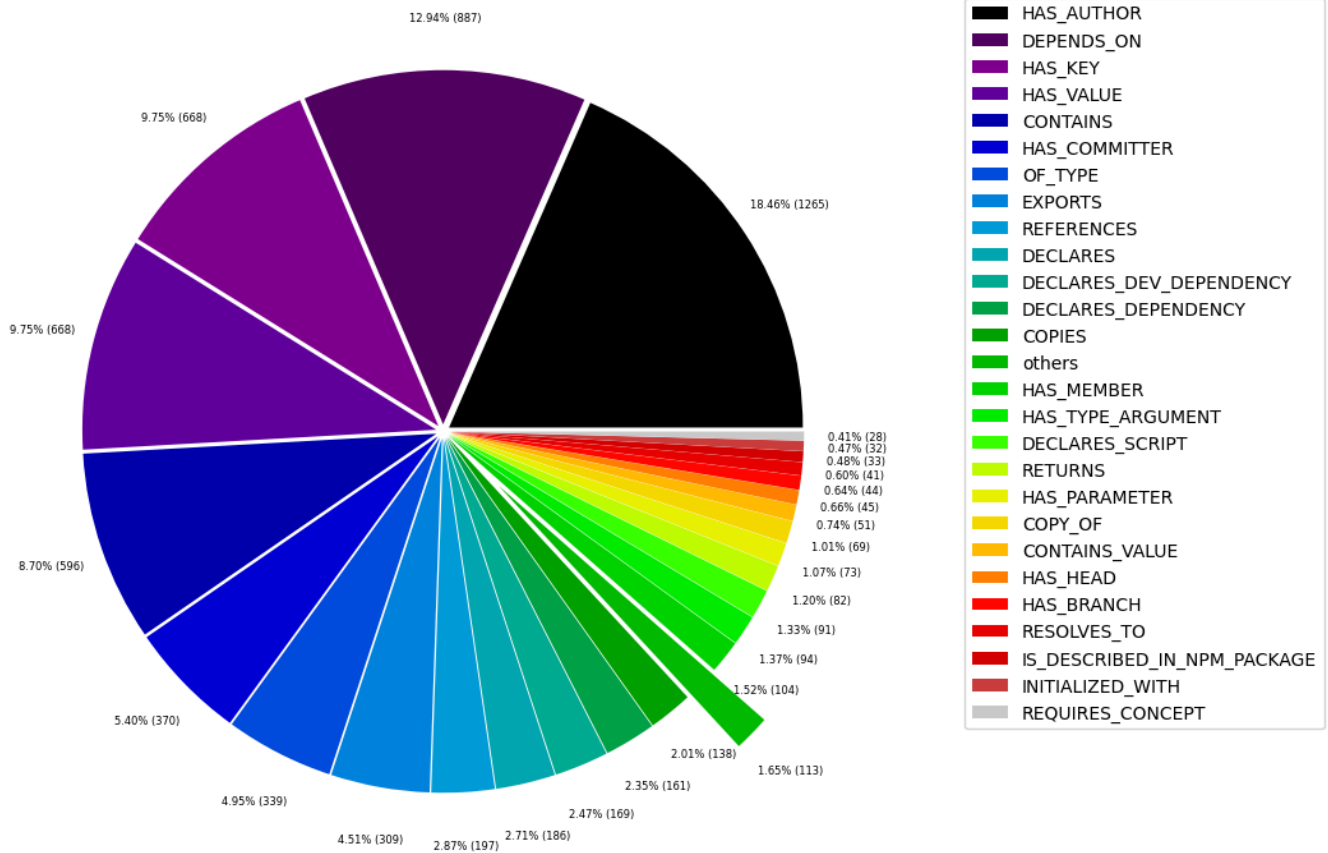
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000299
1	CONSTRAINED_BY	4	0.001196
2	IN_REPOSITORY	5	0.001495
3	REFERENCED_PROJECTS	5	0.001495
4	PARENT	6	0.001794
5	MEMBER	6	0.001794
6	HAS_ROOT	6	0.001794
7	HAS_NPM_PACKAGE	6	0.001794
8	HAS_CONFIG	6	0.001794
9	HAS_ARGUMENT	6	0.001794
10	EXTENDS	6	0.001794
11	DECLARES_ENGINE	6	0.001794
12	CONTAINS_PROJECT	6	0.001794
13	CALLS	6	0.001794
14	DECLARES_PEER_DEPENDENCY	8	0.002392
15	USES	11	0.003289
16	INCLUDES_CONCEPT	19	0.005681
17	REQUIRES_CONCEPT	28	0.008372
18	INITIALIZED_WITH	32	0.009568
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009867
20	RESOLVES_TO	41	0.012258
21	HAS_BRANCH	44	0.013155
22	HAS_HEAD	45	0.013454
23	CONTAINS_VALUE	51	0.015248
24	COPY_OF	69	0.020630
25	HAS_PARAMETER	73	0.021826
26	RETURNS	82	0.024517
27	DECLARES_SCRIPT	91	0.027208
28	HAS_TYPE_ARGUMENT	94	0.028105
29	HAS_MEMBER	104	0.031095

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Update, Change]	MODIFIES	[File, Git]	58394	58394	
1	[Git, Update, Change]	UPDATES	[File, Git]	58394	58394	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	58394	11260	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	17242	11260	
4	[Git, Change, Create]	MODIFIES	[File, Git]	17242	17242	
5	[Git, Change, Create]	CREATES	[File, Git]	17242	17242	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12340	11260	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11260	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11260	1265	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11260	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8758	11260	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	8758	8758	
12	[Git, Change, Delete]	DELETES	[File, Git]	8758	8758	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5891	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3354	11260	
15	[Git, Change, Rename]	DELETES	[File, Git]	3354	3354	
16	[Git, Change, Rename]	RENAMES	[File, Git]	3354	3354	
17	[Git, Change, Rename]	CREATES	[File, Git]	3354	3354	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	3354	3354	
19	[File, Git]	HAS_NEW_NAME	[File, Git]	1770	5891	
20	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1732	1	
21	[Git, Tag]	ON_COMMIT	[Git, Commit]	1732	1732	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1265	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

total_number_of_nodes (vertices): 112006

total_number_of_relationships (edges): 334464

-> total directed graph density: 2.666064678598057e-05

-> total directed graph density in percent: 0.002666064678598057